

Install and Launch

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The project is portable: it is not installed into the system, writes nothing to the registry, and lives entirely inside its own folder. Copying the folder to another machine is a sufficient "install".

Requirements

Windows. The program targets Windows: moving columns by address works through Excel, and the window runs on the system Edge WebView2 engine.

Excel. Needed only by the "Move columns by address" command — it is what preserves the formulas, formatting, and row order of the primary workbook. Every other command works without Excel.

Internet. Needed once to build the environment, and occasionally afterwards to download the OKTMO registry.

A built portable project needs nothing else: Python lives inside, in the `runtime` folder.

Launching

Way	What it does
<code>Start.exe</code>	Open the program window
<code>launcher_gui.cmd</code>	The same
<code>launcher_project.cmd</code>	Project menu: runs, settings, build
<code>launcher_project_ru.cmd</code>	The same menu in Russian
<code>launcher_tools.cmd</code>	Service menu: environment check, licenses, release
<code>builder_main.cmd</code>	Portable environment build

The window launcher looks for Python in this order:

1. `runtime\pythonw.exe`
2. `runtime\python.exe`
3. `runtime\python\pythonw.exe`
4. `runtime\python\python.exe`
5. a system `py -3.12`

The resolved interpreter raises a local server on `127.0.0.1:8080` and opens a window on the Edge WebView2 engine. The window process id is written to `logs/gui_server.pid`.

If the Window Does Not Open

Set `AUDION_GUI_CONSOLE=1` and run the launcher again — it opens in a console and the error messages become visible:

```
set AUDION_GUI_CONSOLE=1 && launcher_gui.cmd
```

Common causes:

- **The environment is not built.** There is no `python.exe` in `runtime` — build the environment.
- **Port 8080 is taken.** The window comes up on another port or not at all; free the port.
- **No WebView2.** The component ships with modern Windows; on older builds it is installed separately. As a fallback, the window opens in a browser.

Building the Environment

Needed only when the project arrived without a `runtime` folder, or when you rebuild it.

`builder_main.cmd` is the build menu:

Item	What it does
[01] PYTHON ENV CMD	Build the portable environment (the main path)
[02] PYTHON ENV PS	The same via PowerShell
[03] FZF	Install or update <code>system_core\fzf.exe</code> for the launcher menus
[09] PORTABLE OFFLINE	Install from the local <code>wheelhouse</code> without network
[70] CLEAN INSTALL CACHE	Clean the build cache; the built runtime survives
[71] VERIFY / DOCTOR	Check the environment and imports
[77] MAKE RELEASE ARCHIVE	Build a release archive

Installing Without Internet

If the project carries a `wheelhouse` folder with dependency wheels, the environment is built from it: `install\install_portable_offline.cmd` or menu item `[09] PORTABLE OFFLINE`.

Verification

`[71] VERIFY / DOCTOR` — or `install\verify_portable_env.cmd` — checks the interpreter and the import of every dependency. It is the first thing to run when something does not work.

Dependencies

Package	What for
<code>pandas</code>	Table operations
<code>openpyxl</code>	Reading and writing XLSX
<code>rapidfuzz</code>	Approximate comparison of street names

Package	What for
<code>pymorphy3</code> , <code>pymorphy3-dicts-ru</code>	Grammatical forms of settlement names
<code>PyMuPDF</code>	PDF text layer
<code>pywin32</code>	Excel automation for column transfer
<code>nicegui</code> , <code>pywebview</code>	The program window
<code>pyyaml</code>	Configuration files
<code>tqdm</code> , <code>rich</code>	Progress and log output
<code>pytest</code> , <code>beautifulsoup4</code>	Tests

License texts of every third-party component live in `licenses/`.

Project Folders

Folder	Contents	Safe to clean
<code>system_core</code>	Source code	No
<code>config</code>	Settings, OKTMO keys and pins	No
<code>data</code>	OKTMO registry and region reference	The registry can be re-downloaded
<code>runtime</code>	Portable Python	Rebuildable
<code>wheelhouse</code>	Wheels for offline install	Yes
<code>input</code>	Source files	Yes
<code>output</code>	Results	Yes
<code>report</code>	Run reports	Yes
<code>logs</code>	Logs	Yes
<code>workspace</code>	Intermediate files	Yes
<code>release</code>	Result archives	Yes

Folder	Contents	Safe to clean
<code>docs</code>	Documentation	No
<code>licenses</code>	Third-party licenses	No
<code>tests</code>	Tests	No

Initial creation of the working folders: `install\init_folders.cmd`.

Maintenance

Project cleanup. `cleanup_project.cmd` removes generated artifacts: caches, temporary conversions, logs, reports, and the rebuildable runtime. Code, settings, documentation, sources, results, and approved references are kept.

CMD file encoding. After editing any `.cmd`, run `install\Check-CmdEncoding.cmd`: the files must stay UTF-8 without BOM with CRLF line endings. Otherwise Cyrillic in the menus turns into garbage.

Dependency updates. `install\Update-Requirements-Lock.cmd` regenerates the lock file.

The OKTMO registry is not updated from here but by a button in the window — see [OKTMO](#).

Before a Large Run

Check the environment (`VERIFY / DOCTOR`), work with copies of important tables, and keep sources, reference, and results in separate folders.

Make the first run on a new dataset over a dozen rows whose answer you know in advance.